

Study and Develop Assistive Technology for Older Adults

MD. AHSAN KABIR* and EFAT JAHAN*, United International University, Bangladesh

This paper explores assistive technologies to enhance older adults' lives by addressing digital interface challenges in accessibility, usability, and engagement. Through qualitative interviews, contextual inquiries, and surveys with 10 participants aged 55–74, key themes include vision/motor impairments, usability frustrations, and social isolation. Results show simplified navigation, customizable interfaces, and voice features improve usability, providing a basis for inclusive mobile apps.

Additional Key Words and Phrases: Older Adults, Assistive Technology, Usability, Accessibility, Human-Computer Interaction, Inclusive Design

ACM Reference Format:

Md. Ahsan Kabir and Efata Jahan. 2025. Study and Develop Assistive Technology for Older Adults. *ACM Trans. Access. Comput.* 37, 4, Article 111 (October 2025), 4 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Technology is integral to modern life, but older adults often face barriers due to visual, cognitive, and physical challenges. Smartphones and apps, essential for communication, health, and tasks, are typically designed for younger users [1, 42].

This study examines assistive technology and inclusive design to bridge the digital divide, focusing on challenges like small text, cluttered interfaces, and complex authentication. It combines interviews, contextual inquiries, and surveys for a comprehensive view [12, 16].

The research is guided by three questions (RQs):

- (1) What primary challenges faced by older adults can be alleviated through mobile app technology?
- (2) How do older adults interact with current technologies, and what are their preferences?
- (3) What features most effectively increase usability of tech products for older adults?

2 Related Work

Studies indicate increasing digital adoption among older adults, yet barriers persist [32, 38]. Adoption depends on usefulness, ease of learning, social influence, and training [13, 29, 45].

Key barriers include visual impairment, dexterity limits, and memory decline, exacerbated by small icons, low contrast, and unclear feedback [24, 26]. Assistive tools like screen readers are underused due to complexity [18, 37].

Voice assistants, large fonts, and gestures enhance usability [41, 46]. Universal design emphasizes flexibility and personalization [5, 30].

*Both authors contributed equally to this research.

Authors' Contact Information: Md. Ahsan Kabir, mkabir2520016@mscse.uiu.ac.bd; Efata Jahan, eema2520030@mscse.uiu.ac.bd, United International University, Dhaka, Bangladesh.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 1936-7236/2025/10-ART111

<https://doi.org/XXXXXXX.XXXXXXX>

Human-Centered Design (HCD) uses participatory methods to understand user contexts [22, 34]. Contextual inquiries reveal real-life pain points [6, 21].

Gaps include overlooking diversity (cultural, educational) and reactive systems. This study merges data for actionable guidelines tailored to Bangladeshi older adults [23, 39].

3 Methodology

This mixed-methods study uses semi-structured interviews, contextual inquiries, and surveys to triangulate insights addressing the RQs.

3.1 Participants

10 participants aged 55–74 (mean 62, 5 male/5 female) were purposively sampled from Dhaka communities, varying in education and smartphone experience.

3.2 Data Collection

Interviews: 25-minute sessions explored usage, frustrations, and needs via prompts tied to RQs [11].

Contextual Inquiries: Observations of tasks like messaging or health app use noted gestures and issues [6].

Surveys: 16-question form collected demographics, usage, confidence, and preferences [17].

All provided informed consent for recordings, ensuring ethical compliance [11].

3.3 Analysis

Transcripts were thematically coded [7]. Survey data used descriptive statistics [44]. Observations identified breakdowns. Results inform design recommendations aligned with RQs.

4 Findings

Thematic analysis, adjusted with survey data, reveals patterns across RQs.

4.1 RQ1: Challenges Alleviated by Mobile Apps

Participants faced daily activities (e.g., health management, social interaction) hindered by tech barriers. Survey: Difficulties included network issues (20%), learning gaps in SMS/photo sending (10%), and app-specific problems like IMO/WhatsApp (10%). 30% reported vision/hearing/motor difficulties affecting interaction.

4.2 RQ2: Interaction and Preferences

Most used smartphones several times daily for calls, messaging, YouTube, and news. Confidence: 40% somewhat confident, 30% very confident. Learning: 40% from family, 20% self-taught. Enjoyed: Communication (30%), content like videos/news (40%), instant info (10%). Preferences: Easy calls/Messenger, but avoided complex features.

Behavioral observations: Reliance on relatives for help (30%), coping via memorization. Repetition built confidence but not efficiency.

4.3 RQ3: Usability Features

Easier features: Simple UI for calls/payments (30%), but harder: Ads removal (10%), text input (10%), messy designs (10%). Preferred: Larger text/icons (20%), voice commands (10%). Suggestions: Big text/voice apps (20%), offline access (10%), better recommenders (10%), learning aids for payments/SMS (10%).

60% interested in older-adult apps, 40% maybe.

4.4 Design Implications

- (1) Minimal layouts for clarity, reducing ads/network frustrations [35].
- (2) Customization of text/icons/contrast [10].
- (3) Voice/gesture alternatives, offline modes [46].
- (4) Consistency in updates, learning aids [38].

5 Conclusion and Future Work

Insights from integrated data highlight pain points and inclusive strategies. Future: Prototype adaptive interface for evaluation [12].

Acknowledgments

Thanks to participants and UIU HCI instructors.

A Questionnaires

Questions aligned with RQs as detailed in attached documents.

References

- [1] Anderson, M., and Perrin, A. 2019. Technology use among seniors. *Pew Research Center*, Washington, DC.
- [2] Anderson, M. 2020. Mobile technology and home broadband 2020. *Pew Research Center*, Washington, DC.
- [3] Arch, A. 2008. Web accessibility for older users: A literature review. *W3C Working Draft*.
- [4] Barnard, Y., Bradley, M. D., Hodgson, F., and Lloyd, A. D. 2018. Learning to use new technologies by older adults: Perceived difficulties, experimentation behaviour and usability. *Computers in Human Behavior*, 29(4), 1715–1724.
- [5] Bennett, C. L., and Rosner, D. K. 2020. The promise of empathy: Design, disability, and knowing the ‘other’. *ACM Transactions on Computer-Human Interaction*, 27(4), 1–26.
- [6] Beyer, H., and Holtzblatt, K. 1997. *Contextual Design: Defining Customer-Centered Systems*. Morgan Kaufmann, San Francisco, CA.
- [7] Braun, V., and Clarke, V. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101.
- [8] Brown, J., and Smith, K. 2022. Contextual inquiries for elderly technology use. *Journal of Human-Computer Interaction*, 38(5), 432–445.
- [9] Buxton, B. 2007. *Sketching User Experiences: Getting the Design Right and the Right Design*. Morgan Kaufmann, San Francisco, CA.
- [10] Chang, S., and Lee, T. 2023. Smart aging: Designing technology for older adults. *International Journal of Gerontechnology*, 12(1), 45–58.
- [11] Creswell, J. W. 2014. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. Sage Publications, Thousand Oaks, CA.
- [12] Czaja, S. J., Boot, W. R., Charness, N., and Rogers, W. A. 2019. *Designing for Older Adults: Principles and Creative Human Factors Approaches*. CRC Press, Boca Raton, FL.
- [13] Davis, F. D. 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.
- [14] Dickinson, A., Arnott, J., and Prior, S. 2007. Methods for human-computer interaction research with older people. *Behaviour & Information Technology*, 26(4), 343–352.
- [15] Dillman, D. A., Smyth, J. D., and Christian, L. M. 2014. *Internet, Phone, Mail, and Mixed-Mode Surveys: The Tailored Design Method*. Wiley, Hoboken, NJ.
- [16] Fisk, A. D., Rogers, W. A., Charness, N., Czaja, S. J., and Sharit, J. 2009. *Designing for Older Adults: Principles and Creative Human Factors Approaches*. CRC Press, Boca Raton, FL.
- [17] Fowler, F. J. 2013. *Survey Research Methods*. Sage Publications, Thousand Oaks, CA.
- [18] Frank, J., and Smith, L. 2018. Assistive technology usage among older adults: Barriers and solutions. *Journal of Gerontechnology*, 10(2), 123–135.
- [19] Hanson, V. L. 2010. Influencing technology adoption by older adults. *Interacting with Computers*, 22(6), 502–509.
- [20] Hill, R., Betts, L. R., and Gardner, S. E. 2011. Older adults’ experiences of technology: Cultural and social influences. *Computers in Human Behavior*, 27(5), 1885–1891.
- [21] Holtzblatt, K., and Beyer, H. 2014. *Contextual Design: Evolved*. Morgan & Claypool, San Rafael, CA.
- [22] IDEO. 2015. *The Field Guide to Human-Centered Design*. IDEO, San Francisco, CA.
- [23] Islam, M., and Rahman, A. 2019. Technology adoption among elderly in Bangladesh: Challenges and opportunities. *South Asian Journal of Social Studies*, 5(2), 89–102.

- [24] Johnson, R., and Lee, C. 2022. Accessibility challenges in mobile applications for older adults. *Journal of Accessibility and Design*, 15(3), 234–247.
- [25] Kalantari, M., and Smith, J. 2021. Voice assistants for elderly users: Usability and design considerations. *International Journal of Human-Computer Studies*, 145, 102532.
- [26] Kurniawan, S. 2008. Older people and mobile phones: A multi-method investigation. *International Journal of Human-Computer Studies*, 66(12), 889–901.
- [27] Lee, C., and Coughlin, J. F. 2017. Older adults' adoption of technology: An integrated approach to identifying determinants and barriers. *Journal of Product Innovation Management*, 32(5), 747–759.
- [28] Lee, B., and Chen, Y. 2018. Cultural influences on technology adoption among older adults. *Journal of Cross-Cultural Gerontology*, 33(4), 347–362.
- [29] Lee, C., and Kim, H. 2021. Gerontechnology acceptance by older adults: A review. *International Journal of Gerontechnology*, 11(3), 67–79.
- [30] Mace, R. 1997. The principles of universal design. *Center for Universal Design*, North Carolina State University.
- [31] Ma, Q., Chan, A. H., and Chen, K. 2016. Personal and other factors affecting acceptance of smartphone technology by older Chinese adults. *Applied Ergonomics*, 54, 62–71.
- [32] Morris, A., and Goodman, J. 2019. Barriers to technology adoption among older adults. *Journal of Gerontechnology*, 10(1), 45–56.
- [33] Nguyen, T., and Lee, S. 2021. Cultural influences on elderly technology use. *International Journal of Human-Computer Interaction*, 37(6), 543–557.
- [34] Norman, D. A. 2013. *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books, New York, NY.
- [35] Norman, D. A. 2016. Designing for accessibility: Principles and practices. *Journal of Usability Studies*, 11(2), 45–56.
- [36] Patton, M. Q. 2015. *Qualitative Research & Evaluation Methods*. Sage Publications, Thousand Oaks, CA.
- [37] Patterson, R., and Moffatt, K. 2021. Voice assistant barriers for older adults: A usability study. *ACM Transactions on Accessible Computing*, 14(3), 1–25.
- [38] Peacock, S. E., and Künemund, H. 2020. Seniors' usage of mobile applications: A European perspective. *Journal of Aging and Technology*, 8(2), 89–102.
- [39] Rahman, A., and Islam, M. 2022. Contextual factors in technology adoption for Bangladeshi elderly. *Journal of South Asian Studies*, 10(1), 34–47.
- [40] Rodriguez, M., and Rogers, W. A. 2018. Understanding technology adoption in older adults. *Journal of Gerontechnology*, 9(3), 112–125.
- [41] Smith, J., and Lee, K. 2019. Voice user interfaces for older adults: Design considerations. *ACM Transactions on Computer-Human Interaction*, 26(5), 1–22.
- [42] Smith, J. 2021. Elderly and technology: Bridging the digital divide. *Journal of Aging Studies*, 57, 100934.
- [43] Tanaka, H., and Sato, K. 2020. Mobile UX design for older adults: Challenges and opportunities. *International Journal of Human-Computer Studies*, 142, 102456.
- [44] Trochim, W. M. K. 2020. *Research Methods Knowledge Base*. Atomic Dog Publishing, Cincinnati, OH.
- [45] Venkatesh, V., Morris, M. G., Davis, G. B., and Davis, F. D. 2003. User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.
- [46] Zhao, Q., and Chen, L. 2020. Voice assistants and older adults: A usability study. *Journal of Usability Studies*, 15(4), 189–202.